NEF ABUNDANCE: FRAMEWORK – JET SEPARATION

1. Jet Separation

Let X be a smooth projective threefold over $\mathbb{C}$ such that $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor on X such that $\mathrm{n}(X, D) = 3$ and $D^3 = 0$.

Setup.

- Fix $\epsilon \in \mathbb{Q}_{>0}$ sufficiently small such that

$$343\epsilon^3 < \tfrac{1}{4}, \qquad 40\epsilon < \tfrac{3}{8}.$$

- Fix an ample Cartier divisor A. For integers $n \geq 1$, put

$$a_n = ((D + A/n)^3)^{1/3}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(D + A/n).$$

  Then P_n is ample and rational Cartier, $r_n \to \infty$, and $P_n^3 \to \epsilon^3$. Indeed, $a_n \to 0$ and $|r_n a_n - \epsilon| \leq a_n/2$ for all sufficiently large n.

- Let $p_i : X \times X \to X$ be the projections, for $i = 1, 2$. Put

$$Z = \mathbb{P}_{X\times X}\big(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D)\big).$$

  Let $\pi : Z \to X \times X$ be the projection. Let $S_1, S_2 \subset Z$ be the two summand sections and set

$$U = Z \setminus (S_1 \cup S_2).$$

- Set $\xi = c_1(\mathcal{O}_Z(1))$. For a positive integer s, put

$$R_{n,s} = (p_1 \circ \pi)^* P_n + (p_2 \circ \pi)^* P_n + s\xi.$$

  Whenever $|kR_{n,s}|$ is used, $k > 0$ is an integer with $kR_{n,s}$ Cartier, equivalently kP_n Cartier.

Jet separation. Let W be a smooth variety, L a Cartier divisor, and $z \in W$. For an integer $j \geq 0$, $|L|$ separates j-jets at z if

$$\forall u \in \mathcal{O}_{W,z}(L) \quad \exists \sigma \in H^0(W, \mathcal{O}_W(L)) \quad \sigma_z - u \in \mathfrak{m}_z^{j+1}\mathcal{O}_{W,z}(L).$$

Here $\mathfrak{m}_z \subset \mathcal{O}_{W,z}$ is the maximal ideal.

Lemma 1.1. *There exists a positive integer s such that, for every $n \gg 0$, $|kR_{n,s}|$ separates $8k$-jets at some $z \in U$ for some integer $k > 0$ with $kR_{n,s}$ Cartier.*

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Proof outline for Lemma 1.1. Keep the Setup notation. On a fourfold slice the divisor is $P_n + s\xi$; pullbacks are understood. All estimates below concern $n \to \infty$, with ϵ, s and the remaining Setup data fixed.

(1) *Volumes.* Lemma 4.1 gives

$$R^7_{n,s} = 140 \int_0^s (P_n + aD)^3 (P_n + (s-a)D)^3 \, da,$$
$$\frac{R^7_{n,s}}{s} \longrightarrow 140\epsilon^6, \qquad \frac{(P_n + s\xi)^4}{s} \longrightarrow 4\epsilon^3.$$

Choose a fixed integer $s > 0$ with $(9 + 56\epsilon)^7 < 70s\epsilon^6$. For sufficiently large n,

$$(9 + 56\epsilon)^7 < R^7_{n,s} < 280s\epsilon^6, \qquad \frac{(P_n + s\xi)^4}{s(4\epsilon)^3} < \frac{1}{8} < 1.$$

The first inequality supplies nonzero full kernels; the last is the slice smallness needed in Step 3.

(2) *Base components.* We need bounds independent of n; the displayed coefficients are one admissible choice, subject to the slice smallness in Step 1. Suppose $\varepsilon(R_{n,s}; z) \leq 8$ for every $z \in U$. The volume lower bound *(uses Step 1)* makes the eight full kernels at levels $9 + 8i\epsilon$, $0 \leq i \leq 7$, nonzero. Lemma 4.8 then gives covering base components $V \subset Z$ and resolutions V^* with

$$0 < R^{\dim V}_{n,s} \cdot V \leq \frac{R^7_{n,s}}{(4\epsilon)^{7-\dim V}}$$
$$< \frac{280s}{4^{7-\dim V}} \epsilon^{\dim V - 1} \quad \text{(uses Step 1)},$$
$$K_{V^*} \cdot R^{\dim V - 1}_{n,s} \leq \frac{7 - \dim V}{4\epsilon} R^{\dim V}_{n,s} \cdot V.$$

For a general projection-fibre component F, $1 \leq \dim F \leq 3$, the same lemma gives the canonical slice estimate and

$$0 < (P_n + s\xi)^{\dim F} \cdot F \leq \frac{(P_n + s\xi)^4}{(4\epsilon)^{4-\dim F}}$$
$$< \frac{8s}{4^{4-\dim F}} \epsilon^{\dim F - 1} \quad \text{(uses Step 1)}.$$

(3) *Projection fibres.* Lemmas 2.1 and 3.8, applied as in Lemmas 4.9–4.11, give $r_n = O_{\epsilon,s}(1)$ for each positive-dimensional fibre case, contradicting $r_n \to \infty$. For a saturated fibre over a point, the contradiction is directly

$$s \leq \frac{(P_n + s\xi)^4}{(4\epsilon)^3} < \frac{s}{8} \quad \text{(uses Step 1)}.$$

Thus Lemma 4.12 leaves threefold correspondences. For a resolution $\rho : Y \to V \subset Z$ and the two induced morphisms $g_i : Y \to X$,

$$0 < (\rho^* R_{n,s})^3 < \frac{280s\epsilon^6}{(4\epsilon)^4}, \qquad K_Y \cdot (\rho^* R_{n,s})^2 \le \frac{1}{\epsilon}(\rho^* R_{n,s})^3,$$
$$\rho^* R_{n,s} - r_n g_i^* D \text{ is nef}, \qquad \deg g_i = O_{\epsilon,s}(1) \quad (i = 1, 2).$$

Here and below, $O_{\epsilon,s}(1)$ means a bound independent of n.

(4) *Correspondences.* A moving branch image $J \subset X$ would satisfy, on a resolution J^*,

$$0 < P_n^2 \cdot J < \frac{280s\epsilon^5}{(4\epsilon)^4}, \qquad K_{J^*} \cdot P_n \le \frac{(P_n^2 \cdot J)^2}{P_n^3}.$$

Since $P_n^3 \to \epsilon^3 > 0$, the lower-dimensional induction and curve bound again give $r_n = O_{\epsilon,s}(1)$, excluding moving branch images. Lemma 4.14 gives finite lists of Stein covers; the graph argument in Lemma 4.15 gives an abelian model with $K_Y + g_1^* D$ big. On a further resolution the same testing class satisfies

$$(K_Y + g_1^* D) \cdot (\rho^* R_{n,s})^2 \le \left(\frac{1}{\epsilon} + \frac{1}{r_n}\right)(\rho^* R_{n,s})^3.$$

The curve bound gives $r_n = O_{\epsilon,s}(1)$, the final contradiction.

(5) *Jets.* Hence some $z \in U$ satisfies $\varepsilon(R_{n,s}; z) > 8$. Serre vanishing on the point blowup gives, for sufficiently large divisible k,

$$H^1\big(Z, \mathcal{O}_Z(kR_{n,s}) \otimes \mathfrak{m}_z^{8k+1}\big) = 0.$$

The ideal-sheaf exact sequence gives the $8k$-jet separation defined in Setup.

Draft status. The proof below is extracted from the existing threefold proof draft. Its lower-dimensional hypotheses, model inputs, and source-dependent arguments have not been independently verified. This note isolates the indicated framework lemma for discussion and checking.

2. Lower-dimensional induction hypotheses

All varieties are over $\mathbb{C}$. For $1 \le \dim Y < 3$, assume the following whenever Y is normal projective and $B \ge 0$ is a rational divisor with (Y, B) klt, and M is nef and rational Cartier:

- If $K_Y + B$ is nef, then it is semiample.
- If $K_Y + B$ is pseudo-effective and $K_Y + B + M$ is nef, then $K_Y + B + M$ is numerically equivalent to a semiample rational Cartier divisor.

These are the log abundance and nef abundance induction hypotheses. Birational reductions use the terminating adjoint programs and effective comparisons of the source framework [3, Sections 2 and 8]. They are model inputs, separate from the two abundance hypotheses.

Lemma 2.1. *Let Y be normal projective with $1 \le \dim Y < 3$, and let $B \ge 0$ be a rational divisor with (Y, B) klt. Let $T = K_Y + B$ be pseudo-effective and M a nef rational Cartier divisor. Assume the two lower-dimensional abundance hypotheses above. If, for some $a \in \mathbb{Q}_{>0}$,*

$$(T + aM) \cdot C > 0$$

for every integral curve through a sufficiently very general point of Y, then $T + cM$ is big for every $c \in \mathbb{Q}_{>0}$.

Proof. Use the M-trivial adjoint reduction of [3, Section 8.2], after clearing the Cartier index of M. It gives a birational contraction $Y \dashrightarrow Y'$ with (Y', B') klt, $T' = K_{Y'} + B'$ pseudo-effective, and M' and $T' + bM'$ nef for some rational $b > 0$. Put $A = T' + bM'$. On a common resolution,

$$p^*T = q^*T' + E, \qquad p^*M = q^*M', \qquad E \ge 0 \quad q\text{-exceptional.}$$

The nef abundance hypothesis gives $A \equiv A_0$ with A_0 semiample. If A were not big, a positive-dimensional general fibre of the morphism defined by A_0 would contain a curve C through a very general point, avoiding $q(\operatorname{Exc} q)$, with $A \cdot C = 0$. Pseudo-effectivity of T' and nefness of M' give $T' \cdot C = M' \cdot C = 0$. Its strict lift contradicts the positivity of $T + aM$. Hence A and $T + bM$ are big. For $c > 0$, choose $0 < \lambda < \min\{1, c/b\}$ and write

$$T + cM = \lambda(T + bM) + (1 - \lambda)T + (c - \lambda b)M.$$

The first summand is big and the other two are pseudo-effective. □

3. Intersection and base-locus estimates

For an integral subvariety V of a smooth variety and an effective Cartier divisor F, $\operatorname{mult}_V(F)$ denotes the multiplicity at a general point of V. If f is a local equation at η_V, then

$$\operatorname{mult}_V(F) = \max\{j \ge 0 : f \in \mathfrak{m}_{\eta_V}^j\}.$$

On a resolution of a subvariety, intersection products use the pullbacks of the indicated restricted Cartier divisors.

Lemma 3.1. *Let X be a smooth projective variety of dimension n, and let D be a nef Cartier divisor. Let $\Lambda \subseteq |D|$ be a nonempty linear system and V an integral component of*

its base locus, of codimension c. If $\mathrm{mult}_V(F) \geq h$ *for every* $F \in \Lambda$*, where* h *is a positive integer, then*

$$h^c D^{n-c} \cdot V \leq D^n.$$

Proof. Intersect successively with c general members of Λ, discarding the components contained in the base locus before each further cut. The remaining components meet the next member properly. Since D is nef, discarding effective components does not increase the degree measured by the remaining powers of D. The resulting codimension-c cycle therefore has D-degree at most D^n.

At η_V, the base ideal is primary for the maximal ideal: this uses that V is a component of the base locus. No discarded component contains η_V. The c general local equations $f_1, \ldots, f_c$ form a parameter ideal I in the regular local ring $R = \mathcal{O}_{X,\eta_V}$ of dimension c. Its local intersection multiplicity is $\mathrm{length}(R/I)$. The hypothesis gives $I \subseteq \mathfrak{m}^h$, so monotonicity of Hilbert–Samuel multiplicity yields

$$\mathrm{length}(R/I) = e(I, R) \geq e(\mathfrak{m}^h, R) = h^c.$$

Thus the coefficient of V is at least h^c. Taking D-degree and using nefness for all remaining effective components proves the inequality. □

Lemma 3.2. *Let* W *be smooth projective,* Θ *an effective rational divisor, and* $V \subset W$ *an integral subvariety of positive dimension. Suppose* (W, Θ) *is lc at* η_V *and has an lc place centred on* V*. Let* $V^\nu \to V$ *be the normalization. There are a finite surjective morphism* $\eta : V' \to V^\nu$*, with* V' *normal, an effective rational divisor* $B_{V'}$*, and the trace* $M_{V'}$ *of a b-nef rational b-divisor such that*

$$\eta^*((K_W + \Theta)|_{V^\nu}) \sim_{\mathbb{Q}} K_{V'} + B_{V'} + M_{V'}.$$

Proof. We use the construction in [1, Lemma 7.1]. The dlt blowup [2, Theorem 2.10] gives

$$p^*(K_W + \Theta) = K_Q + B_Q + E_Q, \qquad (Q, B_Q) \text{ dlt}, \quad E_Q \geq 0,$$

where B_Q is the truncated boundary and the effective surplus E_Q is absent near η_V. Choose a dlt stratum S minimal among those dominating V. Adjunction, followed by adding $E_Q|_S$, gives an effective rational divisor B_S and

$$K_S + B_S \sim_{\mathbb{Q}} (p|_S)^*((K_W + \Theta)|_V).$$

The stratum is not contained in E_Q. Minimality makes (S, B_S) klt over η_V, although it need not be globally lc.

Factor through the normalization and the connected-fibre base: $S \xrightarrow{q} V' \xrightarrow{\eta} V^\nu \to V$, with η finite. The discrepancy-round-up rank condition for q is one: over the generic

point the effective klt boundary contributes zero round-up on strict divisors and allows poles only along exceptional divisors. Its pushforward to normal S is therefore $\mathcal{O}_S$, and connectedness gives rank one. Thus this is a klt-trivial fibration in [5, Definition 3.1]. Its formula is

$$\eta^*((K_W + \Theta)|_{V^\nu}) \sim_{\mathbb{Q}} K_{V'} + B_{V'} + M_{V'},$$

with b-nef moduli divisor M by [5, Theorem 3.5]. The discriminant on this initial base is effective: over a prime divisor, an inverse-image component of multiplicity $m \geq 1$ and boundary coefficient $b \geq 0$ bounds its lc threshold by $(1 - b)/m \leq 1$. A negative threshold still gives a positive discriminant coefficient. Only generic sub-klt singularities are required for this use of the formula. □

Lemma 3.3. *Let W be smooth projective, $\Theta \geq 0$ a rational divisor, and $V \subset W$ an integral subvariety of positive dimension. Suppose (W, Θ) is lc at η_V and has an lc place centred on V. For every nef rational Cartier divisor P on V and resolution $\rho : Y \to V$,*

$$K_Y \cdot (\rho^* P)^{\dim V - 1} \leq (K_W + \Theta)|_V \cdot P^{\dim V - 1}.$$

Proof. Apply Lemma 3.2 and intersect with $\dim V - 1$ copies of the pulled-back P. The effective discriminant and the moduli trace have nonnegative intersection; for the latter, push down from a higher base where the moduli divisor is nef. For compatible canonical Weil cycles,

$$\eta_* K_{V'} = (\deg \eta) K_{V^\nu} + \eta_* R_\eta, \qquad R_\eta \geq 0.$$

Discard ramification and divide by $\deg \eta$. Pushing down K_Y to V^ν changes it only by exceptional cycles, whose testing intersection is zero. No rational Cartier assumption on K_{V^ν} is needed. □

Lemma 3.4. *Let W be smooth projective, P a nef rational Cartier divisor, and $k > 0$ an integer with kP Cartier. Let $\Lambda \subseteq |kP|$ be a nonempty linear system and V an integral base-locus component of positive dimension and codimension a. If $\operatorname{mult}_V(F) \geq h$ for every $F \in \Lambda$, where $h > 0$ is an integer, then there exists $c \in \mathbb{Q}_{>0}$ with $c \leq ak/h$ such that, on a resolution $\rho : Y \to V$,*

$$K_Y \cdot (\rho^*(P|_V))^{\dim V - 1} \leq (K_W + cP)|_V \cdot (P|_V)^{\dim V - 1}.$$

Proof. The generic log canonical threshold of the base ideal satisfies $\lambda \leq a/h$, as seen by blowing up the smooth generic centre. On a log resolution, average sufficiently many general members of the subseries with total coefficient λ. This gives an actual effective boundary $\Theta \sim_{\mathbb{Q}} k\lambda P$ which is lc at the generic point of V and has an lc place centred on V. Lemma 3.3 gives the inequality with $c = k\lambda$. The construction uses actual general members of the given subseries, and requires lc only at η_V. □

Lemma 3.5. *Let W be smooth projective, L a line bundle, and $z : T \to W$ a morphism from a smooth integral variety. Fix integers $m_1 > m_0 \geq 0$ and $1 \leq h \leq m_1 - m_0 + 1$. Suppose the spaces*

$$\mathcal{H}_i(t) = H^0(W, L \otimes \mathfrak{m}_{z(t)}^{m_i}) \quad (i = 0, 1)$$

form vector bundles, and a reduced family $\Gamma \subset T \times W$ dominates W, with each general fibre Γ_t a component of both base loci. Then the higher base ideal satisfies $\mathfrak{b}_{1,t} \subset I_{\Gamma_t}^h$ at a general incidence point.

Proof. A frame of the higher kernel has the form $s(t, w) = \sum_\alpha a_\alpha(t) s_\alpha(w)$ for fixed global sections. A parameter derivative of order $j < h$ loses at most j orders at the moving mark, so belongs to the full lower kernel and vanishes on Γ. At a general smooth point, dominance of $\Gamma \to W$ makes parameter directions span the normal space. Choose transverse parameters t' and write Γ locally as $u = t' - F(t'', w) = 0$. Differentiation in t' with t'', w fixed is differentiation in u. Thus all terms of degree less than h in the normal parameters vanish, giving $s \in I_\Gamma^h$. Faithful flatness of the analytic local ring gives the algebraic containment. Restricting the finite frame to a general parameter fibre proves the assertion, as in [1, Lemma 7.3]. □

Lemma 3.6. *Let $g : W \to B$ be a morphism with W smooth projective, and let $\Lambda \subseteq |L|$ be a nonempty linear system with base ideal $\mathfrak{b}$. Let V be an integral base-locus component and suppose $\mathfrak{b} \subset I_V^h$ near η_V, with $h > 0$ an integer. For general $b \in g(V)$, let F be a reduced component of $V \cap W_b$, where $W_b = g^{-1}(b)$. If W_b is smooth and integral and $F \subsetneq W_b$, then the restricted system on W_b is nonzero, F is a base-locus component, and its base ideal satisfies*

$$\mathfrak{b}\mathcal{O}_{W_b} \subset I_F^h \quad \text{near } \eta_F.$$

Proof. Remove the other base components and work where $V \to g(V)$ is smooth over its actual image. Its scheme fibre is reduced there, hence $I_V \mathcal{O}_{W_b} = I_F$ and $I_V^h \mathcal{O}_{W_b} = I_F^h$. The local base support is exactly F. Since F is proper, not all restricted sections vanish identically on W_b. The resulting base ideal is primary at η_F. This is the actual-image restriction in [1, Lemma 7.4]. □

Lemma 3.7. *Let W be a smooth projective variety of dimension d, and let P be ample and rational Cartier. Fix an integer $k > 0$ with kP Cartier and levels $0 \leq \tau_0 < \cdots < \tau_d$ with common gap $\delta > 0$ and $k\delta \geq 4$. Suppose that, at a very general marked point z, all full spaces*

$$\mathcal{H}_i(z) = H^0(W, \mathcal{O}_W(kP) \otimes \mathfrak{m}_z^{\lceil k\tau_i \rceil})$$

are nonzero, and the lowest base locus has a positive-dimensional component through z. Then there is an algebraic covering family of proper integral components V_t, common to

two adjacent base loci, with $1 \le \dim V_t < d$. *At a general incidence point the entire higher-kernel base ideal satisfies*

$$\mathfrak{b}_t \subset I^h_{V_t}, \qquad h = \lceil k\delta/2 \rceil, \qquad k/h \le 2/\delta.$$

Consequently, on a resolution V_t^*,

$$\begin{aligned} &0 < P^{\dim V_t} \cdot V_t \le (2/\delta)^{d-\dim V_t} P^d, \\ &K_{V_t^*} \cdot P^{\dim V_t - 1} \le (K_W + cP)|_{V_t} \cdot P^{\dim V_t - 1}, \qquad 0 < c \le 2(d - \dim V_t)/\delta. \end{aligned} \tag{1}$$

Proof. On the constant-rank open, the full kernels are vector bundles with nested proper base loci. Track a component through the mark across the $d+1$ levels. Its dimensions lie in $\{1, \dots, d-1\}$, so two adjacent tracked components coincide. After a finite extension of the parameter field, spread them out to an integral family whose marked incidence dominates W. The adjacent orders satisfy

$$\lceil k\tau_{i+1} \rceil - \lceil k\tau_i \rceil \ge k\delta - 1 \ge h - 1.$$

Lemma 3.5 gives the ordinary-power containment for the entire higher base ideal. Apply Lemmas 3.1 and 3.4 for the degree and canonical bounds. The construction follows [3, Lemmas 4.5–4.6]. □

Lemma 3.8. *There are integers* $m_1, m_2, m_3 > 0$ *with the following property. Let* Y *be smooth projective with* $1 \le \dim Y \le 3$. *Let* B_0 *be a reduced simple normal crossing divisor or zero,* L *a nef Cartier divisor of full nef dimension, and* P *a nef rational Cartier divisor. Let* $r, C_0, C_1 > 0$. *If*

$$0 < P^{\dim Y} \le C_0, \quad P - rL \text{ is nef}, \quad K_Y + B_0 + L \text{ is big},$$

and

$$(K_Y + B_0 + L) \cdot P^{\dim Y - 1} \le C_1 P^{\dim Y},$$

then

$$r \le (m_{\dim Y} C_1)^{\dim Y - 1} C_0,$$

Here $m_{\dim Y}$ *depends only on* $\dim Y$.

Proof. Birkar–Zhang [6, Theorem 1.3], with boundary coefficients in $\{0, 1\}$ and Cartier index one for L, makes $|m_{\dim Y}(K_Y + B_0 + L)|$ birational. Resolve its base ideal and let H be the free moving part. Then H is nef and big and

$$H \cdot P^{\dim Y - 1} \le m_{\dim Y} C_1 P^{\dim Y}.$$

Khovanskii–Teissier gives

$$P \cdot H^{\dim Y - 1} \le P^{\dim Y} \left(\frac{H \cdot P^{\dim Y - 1}}{P^{\dim Y}} \right)^{\dim Y - 1} \le (m_{\dim Y} C_1)^{\dim Y - 1} C_0.$$

General hyperplanes through the image of a very general point under the birational morphism defined by H cut an effective curve cycle containing a curve C through that point. Full nef dimension and integrality give $L \cdot C \geq 1$. Thus $r \leq P \cdot C \leq P \cdot H^{\dim Y - 1}$. For $\dim Y = 1$ take $C = Y$. This is [3, Lemma 4.1]; the constants depend only on the dimension. □

4. Proof of Lemma 1.1

4.1. **The projective bundle and adjoint bigness.** In this section, write $D_i = (p_i \circ \pi)^* D$ and $P_i = (p_i \circ \pi)^* P$ for the technical intersection calculations. For $P = P_n$ and $s > 0$, write $R = R_{n,s}$. On a slice over one fixed base coordinate, put

$$Z_{\mathrm{sl}} = \mathbb{P}(\mathcal{O}_X \oplus \mathcal{O}_X(D)), \qquad R_{\mathrm{sl}} = P + s\xi.$$

We use $\pi_* \mathcal{O}_Z(a) = \mathrm{Sym}^a(\mathcal{O}_X(D)_1 \oplus \mathcal{O}_X(D)_2)$; thus $\dim Z = 7$ and $\dim Z_{\mathrm{sl}} = 4$.

Lemma 4.1. *With $P_a = P + aD$, the projective-bundle formulas give*

$$R^7 = 140 \int_0^s P_a^3 P_{s-a}^3 \, da, \qquad R_{\mathrm{sl}}^4 = 4 \int_0^s P_a^3 \, da.$$

For fixed s, these tend to $140 s\epsilon^6$ and $4s\epsilon^3$, respectively, as $n \to \infty$. Also

$$K_Z \sim D_1 + D_2 - 2\xi, \qquad K_{Z_{\mathrm{sl}}} \sim D - 2\xi.$$

Proof. For a split rank-two bundle, $\pi_*(\xi^j) = \sum_{a+b=j-1} D_1^a \cdot D_2^b$ for $j \geq 1$. Expand R^7 and integrate the binomial terms to obtain the first identity; its coefficient is $7!/(3!3!) = 140$. The analogous one-factor expansion gives the second identity. Since $D^3 = 0$, $P^3 \to \epsilon^3$, and $r_n \to \infty$, $(P + aD)^3 / P^3 \to 1$ uniformly for $0 \leq a \leq s$. The two limits follow. The canonical formulas are the rank-two projective-bundle formula with $K_X \sim 0$. □

Lemma 4.2. *Let U, M' be rational Cartier divisors on a normal projective variety, with U semiample and dU Cartier and free for an integer $d > 0$. Let $0 < t_0 < t$ satisfy*

$$(1 - t_0/t)/d > 12 t_0/t.$$

If C generates a $(U + t_0 M')$-negative extremal ray and $(U + tM') \cdot C \geq -12$, then $U \cdot C = 0$. A contraction or flip of such a ray preserves the semiampleness of U.

Proof. If $U \cdot C > 0$, integrality gives $U \cdot C \geq 1/d$, and

$$(U + t_0 M') \cdot C = (1 - t_0/t) U \cdot C + (t_0/t)(U + tM') \cdot C \geq (1 - t_0/t)/d - 12 t_0/t > 0,$$

a contradiction. Thus $U \cdot C = 0$ on every contracted ray. The free line bundle dU descends through a contraction and pulls back through a flip, preserving semiampleness. □

Lemma 4.3. *Let X be a smooth projective threefold with $K_X \sim 0$, and let D be a nef Cartier divisor with $\mathrm{n}(X, D) = 3$. Let $G \geq 0$ be rational with (X, G) klt and $\kappa(G) > 0$. There exist $t_0 > 0$ rational, a normal projective model Y, and a common resolution $p : W \to X$, $q : W \to Y$, with $U = K_Y + G_Y$ semiample, $Q_0 = U + t_0 M'$ nef, and*

$$p^*G = q^*U + E_0, \qquad p^*(G + t_0 D) = q^*Q_0 + E_1, \qquad E_0, E_1 \geq 0 \text{ } q\text{-exceptional.} \tag{2}$$

Moreover, $\kappa(U) = \kappa(G)$, $\mathrm{n}(Y, Q_0) = 3$, and M' is the transform of D, viewed as the trace of its nef b-divisor.

Proof. We spell out [3, Proposition 3.3] in the only threefold instance required below. Run the ordinary G-MMP to a good model; ordinary threefold abundance is [7]. Its adjoint $U = K_{Y_0} + G_{Y_0}$ is semiample. Let M'_0 be the trace of the fixed nef b-divisor determined by D on X. For small $t > 0$, this finite first program is also negative for $G + tD$, and $(Y_0, G_{Y_0} + t\mathbf{D})$ is generalized klt.

Choose dU free and t_0 as in Lemma 4.2. Run a generalized $(U + t_0 M'_0)$-program. The nef data are rational and NQC; ordinary klt approximation gives its steps, and [8, Theorem 1.2] gives termination. The generalized length bound is 12 in this threefold setting [3, Lemma 3.1]. Lemma 4.2 makes every step U-trivial. On the final model Y, U remains semiample, $Q_0 = U + t_0 M'$ is nef, and a common resolution gives the two displayed comparisons.

Moreover Q_0 has full nef dimension. Otherwise a very general compact nef-reduction fibre has a curve through its very general point avoiding $q(\operatorname{Exc} q)$, which has codimension at least two. Nefness of U, pseudo-effectivity of M', and generic nonnegativity of pseudo-effective degrees give $U \cdot C = M' \cdot C = 0$. Lifting this curve in both comparisons gives D-degree zero through a very general point of X, a contradiction. □

Lemma 4.4. *Let X be a smooth projective threefold with $K_X \sim 0$, and let D be a nef Cartier divisor with $\mathrm{n}(X, D) = 3$. If $G \geq 0$ is rational, (X, G) is klt, and $\kappa(G) > 0$, then $G + cD$ is big for every rational $c > 0$.*

Proof. Use the model in Lemma 4.3. The contraction $h : Y \to S$ defined by U has $\dim S = \kappa(G) > 0$ and $U \sim_{\mathbb{Q}} h^*A_S$ with A_S ample. On a very general positive-dimensional fibre F,

$$K_F + G_Y|_F \sim_{\mathbb{Q}} 0, \qquad Q_0|_F = t_0 M'|_F \text{ is nef of full nef dimension.}$$

Its dimension is at most two, so Lemma 2.1 makes $Q_0|_F$ big. For zero-dimensional fibres relative bigness is automatic. Relative bigness follows on the generic fibre: for a fixed ample H, one of the countably many closed section loci for $(\ell Q_0 - H)|_{F_t}$ contains a very general point, hence is the whole shrunken flat base. Adding a sufficiently large multiple

bU now makes $Q_0 + bU$ big globally. The comparisons imply that $(1+b)G + t_0 D$ is big. Finally, for $0 < \lambda < \min\{1/(1+b), c/t_0\}$,

$$G + cD = \lambda((1+b)G + t_0 D) + (1 - \lambda(1+b))G + (c - \lambda t_0)D$$

is big. □

Lemma 4.5. *Let X be a smooth projective threefold with $K_X \sim 0$, and let D be nef Cartier with $\mathrm{n}(X, D) = 3$. Let P be ample and rational Cartier, and let $s > 0$ be an integer. Put $R_{\mathrm{sl}} = P + s\xi$ on the quotient projective bundle below. Let $F \subset \mathbb{P}(\mathcal{O}_X \oplus \mathcal{O}_X(D))$ be an integral multisection of degree $e > 0$ meeting the torus open. Writing p for the bundle projection and S_1, S_2 for the summand sections, set $G_i = p_*(S_i \cdot F)$. These are effective integral divisors G_1, G_2 on X satisfying*

$$G_1 - G_2 \sim \pm eD, \qquad sG_i \cdot P^2 \le R_{\mathrm{sl}}^3 \cdot F. \tag{3}$$

For $G = G_1 + G_2$ and every sufficiently small rational $b > 0$, (X, bG) is klt, $\kappa(bG) > 0$, and $bG + D$ is big.

Proof. The norm of the ratio of the two fibre coordinates gives the linear equivalence. If S is either axis, $R_{\mathrm{sl}} - sS$ and $R_{\mathrm{sl}} - P$ are nef. Intersecting these classes against the effective cycle $S \cdot F$ gives the degree inequalities.

If $\kappa(bG) = 0$, ordinary threefold abundance gives a good $(K_X + bG)$-model Y with $K_Y + bG_Y \sim_{\mathbb{Q}} 0$. The transform of K_X is linearly trivial, so $G_Y = 0$: every component of G has been contracted. On a common resolution $a : U \to X$, $c : U \to Y$, the signed relation in (3) expresses $a^*D \sim_{\mathbb{Q}} E$, with E c-exceptional. Divisors exceptional over X are also exceptional over Y, since this MMP extracts none. Since a^*D is nef, negativity gives $E \le 0$. Its class is pseudo-effective, so intersection with an ample square forces $E = 0$. This contradicts full nef dimension. Thus $\kappa(bG) > 0$, and Lemma 4.4 gives bigness. The axis construction and negativity argument are dimension-independent; only the ordinary good-model input is being specialized to threefolds. □

Lemma 4.6. *Let T_1, T_2 be non-uniruled integral projective varieties over $\mathbb{C}$ of the same dimension. If an integral algebraic family of graphs of birational maps $T_1 \dashrightarrow T_2$ dominates $T_1 \times T_2$, both varieties are birational to an abelian variety.*

Proof. Identify the two varieties by one map in the family. Hanamura's theorem, in the form [9, Theorem 3.8], supplies a smooth projective birational model Y with $\mathrm{Bir}(Y)_{\mathrm{red}}$ a group scheme locally of finite type. Its identity component is $\mathrm{Aut}^0(Y)$, an abelian variety acting regularly.

Conjugate the family to Y, take the closure of its relative graphs, and shrink the integral parameter space so that these graphs are flat and geometrically integral, with

both projections birational. This shrinking is necessary: graph flatness is not preserved automatically by birational conjugation [9, Remark 3.6]. The flat-graph representability theorem [9, Proposition 3.7] then gives a morphism from the reduced parameter variety to $\mathrm{Bir}(Y)_{\mathrm{red}}$. Its connected image lies in one translate of the identity component. After translating by a fixed member, product dominance implies that a general point has a dense orbit under that abelian group. Such an orbit is the quotient by its closed stabilizer, hence an abelian variety. It is therefore a birational model of Y. □

Lemma 4.7. *Let $p : Y \to A_0$ be a birational morphism from a smooth projective variety to an abelian variety. If L is nef Cartier of full nef dimension on Y, that is, $\mathrm{n}(Y, L) = \dim Y$, then $K_Y + L$ is big.*

Proof. The numerical pushforward $M' = p_*L$ is pseudo-effective on A_0, thus nef. If it were not big, the kernel of the semipositive Hermitian form of its integral class would define a positive-dimensional abelian subvariety on whose translates M' is numerically trivial. The image of the exceptional locus of p has codimension at least two. A general translate meets that image in codimension at least two; complete-intersection curves in the translate can be chosen through a very general point avoiding it. Their strict lifts have L-degree zero, a contradiction. Thus M' is big.

Negativity gives $p^*M' - L = E_0 \geq 0$, with E_0 exceptional. Since A_0 is smooth and $K_{A_0} \sim 0$, $K_Y \sim E_K \geq 0$, with positive coefficient along every p-exceptional divisor. Choose $0 < u < 1$ rational with $E_K - uE_0 \geq 0$. Then

$$K_Y + uL \sim_{\mathbb{Q}} up^*M' + (E_K - uE_0)$$

is big. Adding $(1-u)L$ proves the assertion. This lemma has no dimension restriction. □

4.2. **Base components and slice exclusions.** Fix $\delta = 8\epsilon$, $\tau_i = 9 + i\delta$ for $0 \leq i \leq 7$, and choose s so that $\tau_7^7 < 70s\epsilon^6$. For all sufficiently large n, Lemma 4.1 gives

$$\epsilon^3/2 < P^3 < 2\epsilon^3, \qquad \tau_7^7 < R^7 < 280s\epsilon^6, \qquad R^4_{\mathrm{sl}} < 8s\epsilon^3.$$

In particular $(2/\delta)^3\, 8\epsilon^3 = 1/8 < 1$. For the following lemmas, these choices remain fixed, and $C_0 = (2/\delta)^4\, 280s\epsilon^6$.

Lemma 4.8. *For all sufficiently large n, if $\varepsilon(R; z) \leq 8$ at every point of the torus open, there is a covering family of proper integral $V \subset Z$ such that*

$$0 < R^{\dim V} \cdot V \leq (2/\delta)^{7-\dim V} R^7, \qquad K_{V^*} \cdot R^{\dim V - 1} \leq \frac{2(7 - \dim V)}{\delta} R^{\dim V} \cdot V. \tag{4}$$

For $i = 1, 2$, set $\pi_i = p_i \circ \pi$. For a general reduced fibre component F of $V \to \pi_i(V)$, viewed in the opposite fourfold slice, if $1 \leq \dim F = \dim F \leq 3$, the actual restricted

system gives

$$
\begin{aligned}
&0 < R_{\mathrm{sl}}^{\dim F} \cdot F \le (2/\delta)^{4-\dim F} R_{\mathrm{sl}}^4,\\
&K_{F^*} \cdot R_{\mathrm{sl}}^{\dim F-1} \le (K_{Z_{\mathrm{sl}}} + c_{\dim F} R_{\mathrm{sl}})|_F \cdot R_{\mathrm{sl}}^{\dim F-1}, \quad 0 < c_{\dim F} \le 2(4-\dim F)/\delta.
\end{aligned} \tag{5}
$$

Both images of a general marked incidence point are very general on X.

Proof. Asymptotic Riemann–Roch and $\tau_7^7 < R^7$ make the eight full kernels nonzero for large divisible k. A curve through a very general z with R-degree divided by multiplicity less than τ_0 is contained in the lowest base locus, by the intersection multiplicity inequality. Hence that base locus is positive-dimensional. Lemma 3.7 gives the covering family and the first bounds. Since K_Z is minus the axes and V meets their complement, the ambient canonical contribution is nonpositive. Apply Lemma 3.6 to either projection over its actual image, and then apply Lemmas 3.1 and 3.4 in the smooth fourfold slice. This gives the fibre bounds. □

For the next three lemmas, F is a general reduced fibre component of $V \to \pi_i(V)$ in a covering family as in Lemma 4.8; W_0 is its image in the other copy of X. We require $1 \le \dim F \le 3$, the slice estimates (5), and the stated very-general-image conditions.

Lemma 4.9. *For all sufficiently large n, a slice component F satisfying the above conditions cannot be generically finite onto W_0 with* $\dim W_0 < 3$.

Proof. If $F \to W_0$ is generically finite and $\dim W_0 < 3$, the pullback L of D is nef Cartier of full nef dimension on F^*, and K_{F^*} is pseudo-effective by the very-general-images argument. Lemma 2.1 gives $K_{F^*} + L$ big. The testing class $P_Y = R_{\mathrm{sl}}|_{F^*}$ satisfies $P_Y - r_n L$ nef, and (5) gives uniform volume and canonical ratios. Lemma 3.8 bounds r_n, a contradiction. □

Lemma 4.10. *For all sufficiently large n, a slice component F satisfying the above conditions cannot be the full projective-line bundle over W_0 with* $\dim W_0 \le 2$.

Proof. If $F \to W_0$ has one-dimensional fibres, F is the entire projective-line bundle over W_0. Put $w = \dim W_0 = \dim F - 1$. For $w = 0$, (5) gives

$$s = R_{\mathrm{sl}} \cdot F \le (2/\delta)^3 R_{\mathrm{sl}}^4 < s.$$

For $w = 1, 2$, resolve W_0 and take its induced bundle $p_0 : F^* \to W_0^*$. Cancelling the relative canonical term $-2\xi + p_0^* D$, which equals the ambient canonical restriction, gives

$$p_0^* K_{W_0^*} \cdot R_{\mathrm{sl}}^w \le c_{\dim F} R_{\mathrm{sl}}^{w+1} \cdot F.$$

Its term linear in ξ is $wsK_{W_0^*} \cdot P^{w-1}$; all other terms are nonnegative because $K_{W_0^*}$ is pseudo-effective and P, D are nef. For $r_n \geq s$, expansion gives

$$\begin{aligned} (w+1)sP^w \cdot W_0 \leq R_{\mathrm{sl}}^{w+1} \cdot F \leq 7sP^w \cdot W_0, \\ K_{W_0^*} \cdot P^{w-1} \leq (7c_{\dim F}/w)P^w \cdot W_0. \end{aligned} \tag{6}$$

The first lower bound and (5) bound $P^w \cdot W_0$ uniformly. Applying Lemmas 2.1 and 3.8 on W_0^* is again contradictory. □

Lemma 4.11. *For all sufficiently large n, a slice component F satisfying the above conditions cannot be a multisection over X.*

Proof. Lemma 4.5 gives $G = G_1 + G_2$, $bG + D$ big, and $G \cdot P^2 \leq 2R_{\mathrm{sl}}^3 \cdot F/s$. On a log resolution $a : Y \to X$, let B_0 be the reduced strict support of G together with every exceptional divisor. Decrease b so that

$$K_Y + B_0 + a^*D \sim_{\mathbb{Q}} a^*(bG + D) + E_b, \qquad E_b \geq 0.$$

This adjoint is big. Testing with $P_Y = a^*P$, all exceptional divisors contribute zero, and

$$(K_Y + B_0 + a^*D) \cdot P_Y^2 \leq G \cdot P^2 + D \cdot P^2 \leq C, \qquad D \cdot P^2 \leq P^3/r_n.$$

The positive lower bound on P^3 gives a uniform adjoint ratio. Lemma 3.8, with this reduced boundary, excludes the case. □

Lemma 4.12. *For all sufficiently large n, both projections of a general V in Lemma 4.8 are generically finite and dominant. Let $\rho : Y \to V$ be a smooth resolution, put $g_i = \pi_i \circ \rho : Y \to X$ and $P_Y = \rho^*(R|_V)$. Then*

$$0 < P_Y^3 \leq C_0, \qquad K_Y \cdot P_Y^2 \leq (8/\delta)P_Y^3, \qquad P_Y - r_n g_i^* D \text{ is nef.} \tag{7}$$

Moreover, $\deg g_i \leq 2C_0/\epsilon^3$.

Proof. Lemmas 4.9, 4.10, and 4.11 exclude fibre dimensions one, two, and three. A four-dimensional fibre makes V the full inverse image of a proper $W_0 \subset X$, with $\dim W_0 \leq 2$; the other projection then has fibre dimension $\dim W_0 + 1$, already excluded. Thus both projections are generically finite and $\dim V \leq 3$. For $\dim V < 3$, (4) and Lemmas 2.1 and 3.8 give a uniform bound on r_n, a contradiction. Hence $\dim V = 3$. The displayed estimates follow from (4) and nefness of $R - r_n D_i$. Since $R - P_i$ is nef, $\deg g_i \, P^3 \leq P_Y^3 \leq C_0$. □

4.3. **Finite covers and the final exclusion.**

Lemma 4.13. *For all sufficiently large n, no relative Jacobian component in the family of Lemma 4.12 whose general image is a divisor has images sweeping X.*

Proof. Suppose that the images of one branch-divisor component sweep X, and let $J \subset X$ be a general such image. Pulling back a nowhere-vanishing three-form from X gives an effective canonical divisor on Y, containing a ramification divisor over J with positive integral coefficient. Thus

$$0 < d_J := P^2 \cdot J \leq K_Y \cdot P_Y^2 \leq (8/\delta)C_0.$$

Generic numerical subadjunction for (X, J) and Hodge index give on a resolution J^*

$$K_{J^*} \cdot P \leq J^2 \cdot P \leq \frac{(J \cdot P^2)^2}{P^3} \leq C_2 d_J, \qquad C_2 = 16C_0/(\delta\epsilon^3). \tag{8}$$

Here (X, J) is lc at η_J; global lc is unnecessary. Hodge index applies to J without assuming J nef. Since the images sweep, J^* is non-uniruled and the pulled-back D has full nef dimension. The testing volume is d_J, bounded above, its canonical ratio is bounded by C_2, and $P - r_n D$ is nef. Lemmas 2.1 and 3.8 give a uniform bound for r_n. Hence sweeping branch divisors are impossible for large n. □

Lemma 4.14. *Let X be a smooth projective complex variety, $U \subset X$ a nonempty Zariski open subset, and $e > 0$ an integer. Up to isomorphism over X, there are only finitely many finite surjective morphisms $h : T \to X$ with T normal and integral,* $\deg h \leq e$, *and $h^{-1}(U) \to U$ étale.*

Proof. The fundamental group of U is finitely generated, hence has only finitely many subgroups of index at most e. These classify the connected finite étale covers of U. Each normal cover of X is uniquely determined by its function field, as the normalization of X in that field. The assertion follows; compare [1, Lemma 7.5]. □

For a fixed family in Lemma 4.12, Lemma 4.13 puts the finitely many relative divisorial branch images in a proper closed subset of X. Remove this subset and shrink the parameter space. Purity makes each finite normal Stein cover étale over the resulting fixed open U. The degree bound and Lemma 4.14 therefore give finite lists of Stein covers for both projections.

Lemma 4.15. *Let X be a smooth projective threefold with $K_X \sim 0$, and let D be a nef Cartier divisor with* $\mathrm{n}(X, D) = 3$. *Consider an algebraic family of smooth projective threefolds Y_t with generically finite morphisms $g_{i,t} : Y_t \to X$, for $i = 1, 2$, whose product images cover a dense subset of $X \times X$. Suppose that the normal Stein covers of each $g_{i,t}$*

belong to a finite list. Let $r, C_0, C_1 > 0$. If there are nef rational Cartier divisors P_t such that, for every general t,

$$0 < P_t^3 \le C_0, \qquad K_{Y_t} \cdot P_t^2 \le C_1 P_t^3, \qquad P_t - r g_{i,t}^* D \text{ is nef} \quad (i = 1, 2),$$

then

$$r \le \max\{1, m_3^2 (C_1 + 1)^2 C_0\},$$

where m_3 is the integer in Lemma 3.8.

Proof. Each member is birational to both of its Stein covers. Fix T_1, T_2 occurring together in an original member with the displayed bounds. The projected incidence dominates $X \times X$. The possible graph families for these finitely many choices belong to countably many Hilbert schemes; stratify for flat integral fibres and birational projections. Uncountability of $\mathbb{C}$ implies that one such integral graph family has evaluation dominant over $X \times X$. Its evaluation is also dominant on $T_1 \times T_2$: this product is integral, finite over $X \times X$, and of the same dimension.

The covers are non-uniruled, since they are finite over X. Lemma 4.6 makes them birational to an abelian variety A_0. Return to an original member with these covers and resolve further to obtain $p : Y \to A_0$. The bounds are preserved, since exceptional canonical divisors have zero intersection with the pulled-back testing square. Write P for the pulled-back testing divisor. The divisor $L = g_1^* D$ remains nef Cartier of full nef dimension. Lemma 4.7 gives $K_Y + L$ big, while

$$(K_Y + L) \cdot P^2 \le (C_1 + 1/r) P^3.$$

If $r < 1$, the asserted bound is immediate. Otherwise Lemma 3.8, with $C_1 + 1/r \le C_1 + 1$, gives $r \le m_3^2 (C_1+1)^2 C_0$. The sizes of the cover lists do not enter this numerical bound. □

Proposition 4.16. *There is a fixed integer $s > 0$ such that, for all sufficiently large n, some point z in the torus open satisfies $\varepsilon(R; z) > 8$.*

Proof. Use the fixed choices above. If the assertion failed for arbitrarily large n, Lemmas 4.8 and 4.12 would produce bounded threefold correspondences. Resolve and shrink the family, and spread out the finitely many generic Jacobian components after a finite parameter extension. Lemmas 4.13 and 4.14 give finite lists of Stein covers. Apply Lemma 4.15 with $r = r_n$ and $C_1 = 8/\delta$. Its bound contradicts $r_n \to \infty$. □

Proof of Lemma 1.1. Choose z as in Proposition 4.16, and rational $8 < a < \varepsilon(R; z)$. On the blowup $b : \widetilde{Z} \to Z$, $b^* R - a J_z$ is ample. For large divisible k_0, Serre vanishing makes restriction surjective modulo $\mathfrak{m}_z^{k_0 a}$, with $k_0 a \ge 8k_0 + 1$. Thus $|k_0 R_{n,s}|$ separates $8k_0$-jets at z. □

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